

Long and short narrow pore models for membrane polarization

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ABSTRACT

Many recent approaches to derive hydraulic soil and rock parameters from spectral induced polarization data are based on the determination of the characteristic time of measured impedance spectra. The characteristic time, sometimes also termed *relaxation time*, depends on the characteristic length scale of the pore geometry. One commonly used theory describing the complex electrical impedance of porous media is the membrane polarization. The pore space is modeled as a sequence of wide and narrow pores with different anion and cation mobilities. A question of particular interest is the dependence of the characteristic time on the pore lengths. The dominance of diffusion processes leads to a quadratic dependence of the characteristic time on the length scale of the pore system. However, analytical and numerical studies revealed an inconsistency concerning the pore length that determines the characteristic time.

Whereas the analytical short narrow pore (SNP) model predicts a dependence on the length of the narrow pore, the length of the wide pore dominates in various numerical studies. We propose a new SNP model, which involves fewer assumptions concerning the ionic mobilities. Our new SNP model is a special case of the Cole-Cole impedance. Its characteristic time is controlled by the length of the narrow pore. A second model is derived for the case of long narrow pores and termed the *long narrow pore* model. The resulting impedance compares, but is not equal to a special case of the Cole-Davidson impedance and its characteristic time varies with the length of the wide pore. Both new models are special cases of the Marshall-Madden impedance for membrane polarization. The limitation of their respective validity with respect to certain combinations of mobilities and pore lengths, allows for the coexistence of both behaviors and resolves the apparent contradiction between theoretical and numerical studies.

INTRODUCTION

Induced polarization (IP) is a geophysical method based on the frequency-dependent complex resistivity of the near subsurface. Depending on the composition of the subsurface, different polarization mechanisms give rise to a low-frequency dispersion (typically between 1 mHz and 1 kHz) of the complex resistivity (or IP effect). In rocks with metallic constituents, electrode polarization causes a pronounced IP effect (e.g., Bleil, 1953; Pelton et al., 1978; Wong, 1979). With the appearance of more precise data acquisition systems, new applications beyond ore exploration emerged: Today, even small phases (below 1 mrad) caused by clayey sediments, sands, and microfractured rocks can be measured. In these cases, IP signatures are primarily caused by polarization mechanisms related to the electrical double layer (EDL) at the mineral-fluid interface (e.g., Schwarz, 1962; Vinegar and Waxman, 1984), giving rise to new

applications in environmental and hydrogeophysics. For instance, the specific surface area shows a strong correlation with the imaginary part of the complex conductivity (e.g., Börner and Schön, 1991; Slater et al., 2006; Kruschwitz et al., 2010; Weller et al., 2010b) offering a geophysical access to this soil parameter. Because hydraulic permeability also depends on the specific surface area, it may be related to the imaginary part of the complex conductivity by means of empirical equations (e.g., Pape et al., 1987; Börner et al., 1996; Slater and Lesmes, 2002). A similar approach uses the characteristic time scale of the measured resistivity spectrum to derive the hydraulic permeability (e.g., Binley et al., 2005; Kemna et al., 2005; Revil and Florsch, 2010). Therefore, spectral induced polarization (SIP) seems to be a promising method for noninvasive mapping and sounding of hydraulic soil and rock parameters also at the field scale (e.g., Hoerdt et al., 2009). Because of its sensibility with respect to variations in pore fluid chemistry (e.g., Angoran and Madden, 1977; Weller et al.,

Manuscript received by the Editor 30 December 2012; revised manuscript received 29 May 2013; published online 7 October 2013.

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2011) and to geometric alterations of the pore space caused by different contaminants, SIP has also been used for the delineation and characterization of contaminant plumes (e.g., Vanhala, 1997; Schmutz et al., 2010; Flores Orozco et al., 2012).

To reduce the information in the full spectra to a few meaningful parameters, empirical models have been used in data processing and interpretation. The most universal of these models — expressed in terms of a frequency-dependent impedance — is the generalized Cole-Cole model (Pelton et al., 1978), also known as *Havriliak-Negami relaxation* (Havriliak and Negami, 1967)

$$Z_{GCC}(\omega) = Z_{DC} \left[1 - \eta_0 \left(1 - \frac{1}{(1 + (i\omega\tau_{GCC})^c)^a} \right) \right]. \quad (1)$$

Here, ω is the angular frequency, Z_{DC} the low-frequency limit of the impedance, η_0 the polarizability (also termed *chargeability*), and τ is the time constant. The generalized Cole-Cole model, equation 1, is equivalent to the Cole-Cole model (Cole and Cole, 1941), if the exponent a equals unity, and to the Cole-Davidson model, if the exponent c equals unity. Generalized Cole-Cole models or superpositions of these models are apt to fit a wide range of experimental data, but provide little information about the underlying physical processes.

To overcome the shortcoming of the empirical models, and to understand the physical basics of the IP-effect in absence of metallic constituents, mainly two pore-scale theories have been discussed in literature. The grain-based model developed by O'Konski (1960), Schwarz (1962), and Schurr (1964) for dispersed spherical mineral particles describes the polarization of the Stern layer, a thin layer of ions directly adsorbed to the mineral surface. This model has recently been extended by Leroy et al. (2006, 2008) and Leroy and Revil (2009) who incorporated a detailed electrochemical description of the EDL and successfully predicted the impedance of granular media. The second model attributes the IP effect to the membrane-like behavior of narrow passages or clay particles distributed along the pore channel. Membrane polarization was first studied by Marshall and Madden (1959) for a pore space model consisting of a sequence of alternating zones. The zones differ in their length, and to account for the membrane-like behavior, reduced transport numbers (product of ion mobility and concentration) are assigned to the zones representing narrow passages. Subsequently, the pore space based model was modified to include the influence of the pore cross section (Fridrikhsberg and Sidorova, 1961) and to allow for a quantitative estimation of the influence of the EDL on membrane polarization (Buchheim and Irmer, 1979).

Motivated by the strong link between soil hydraulic parameters and SIP measurements, more recent studies have focused on the relation between the characteristic time and pore dimensions (i.e., length and diameter). Starting from the original membrane polarization model, Titov et al. (2002) developed an impedance model for the special case of very short narrow pores (SNP). One of their findings is that the characteristic time shows a quadratic dependence on the length of the narrow passages. On the other hand, based on numerical simulations Blaschek and Hördt (2009) and Volkmann and Klitzsch (2010) report a quadratic dependence on the length of the wide pore.

The small number of existing theoretical studies on membrane polarization is in contrast to a high demand for a better understanding of the underlying physical processes, necessary to profoundly

interpret field and laboratory data. The objective of this study is to clarify the apparently contradicting results of the aforementioned studies concerning the relation between characteristic time and pore length and at the same time provide a deeper insight into the physics of membrane polarization.

In the first section, we briefly review the basics of membrane polarization theory as described by Marshall and Madden (1959) for a 1D pore system. We express the impedance formula of these authors in terms of DC-limit, polarizability, and two time constants, one for each zone. A parameter study varying the lengths of the two zones reveals two clearly distinguishable regimes with respect to the characteristic time-pore length relation.

In the section that follows, we analytically derive two new approximate impedance models, which allow for the analytic determination of their characteristic times. The assumption of long narrow pores leads to an impedance model with a characteristic time that is determined solely by the length of the wide pore. This finding is in accordance with the results of Blaschek and Hördt (2009) and Volkmann and Klitzsch (2010). The assumption of SNP, on the other hand, results in a model with a quadratic dependence of the characteristic time on the length of the narrow pore as reported earlier by Titov et al. (2002).

In the last section, we provide a quantitative criterion for the distinction between the two characteristic time regimes and provide a equivalent circuit representation of the Marshall-Madden impedance involving finite-length Warburg impedances. We also discuss the practical significance of the two new impedance models.

MEMBRANE POLARIZATION THEORY

In the following, we briefly review the mathematical formulation of membrane polarization by Marshall and Madden (1959). An external electrical field accelerates the ions of the electrolytic pore fluid corresponding to their respective charges. In presence of a concentration gradient, this electrically excited migration current superimposes a diffusion current. As no ions are generated or removed, continuity equations for both ion species complete the system of partial differential equations, which describes the electrical current densities (for univalent cations and anions)

$$\mathbf{j}_p = -FD_p \nabla c_p + F\mu_p c_p \mathbf{E}, \quad (2)$$

$$\mathbf{j}_n = -FD_n \nabla c_n - F\mu_n c_n \mathbf{E}, \quad (3)$$

$$F \frac{\partial c_p}{\partial t} = -\nabla \mathbf{j}_p, \quad (4)$$

$$F \frac{\partial c_n}{\partial t} = -\nabla \mathbf{j}_n. \quad (5)$$

Here, $\mathbf{j}_i$ are the electrical current densities in C/(m²s), D_i the diffusion coefficients in m²/s, μ_i the ion mobilities in m²/(Vs), c_i the ion concentrations in mol/m³, F is the Faraday constant (96,485 C/mol), and $\mathbf{E}$ the electrical field in V/m. The subscript p denotes the corresponding quantities of the cations and the subscript n those of the anions. Diffusion coefficients and mobilities are related by the Nernst-Einstein equation

$$D = \frac{\mu k_B T}{e}, \quad (6)$$

where k_B is the Boltzmann constant ($1.38 \cdot 10^{-23}$ J/K), e the unit charge ($1.60 \cdot 10^{-19}$ C), and T the absolute temperature in K. Equations 2–5 are coupled via the Poisson equation, as the buildup of a concentration difference between ion species of opposite charge causes a nonvanishing charge density

$$\Delta\psi = -\nabla\mathbf{E} = -\frac{F}{\varepsilon}(c_p - c_n). \quad (7)$$

Here, ψ denotes the electrical potential in V, $\varepsilon = \varepsilon_0 \varepsilon_r$ the permittivity of the pore fluid, where ε_0 is the vacuum permittivity ($8.854 \cdot 10^{-12}$ C/(Vm)) and ε_r the relative permittivity (≈ 80 for water).

To describe the processes within a complex pore system, the pore system is reduced to an alternating sequence of two interconnected pore types (see Figure 1). Wide pores and narrow pores (pore throats) differ with respect to their lengths ΔL_i and cross sections. At the interface between mineral matrix and pore fluid, (usually) negative charges at the mineral surface cause an EDL to build up. The smaller the pore radius, the bigger the portion of the pore cross section, which is covered by the EDL. The asymmetric ion concentrations in the EDL cause an increase of the cationic and a reduction of the anionic contribution to the total electrolytic current through the pore, which is accounted for by different transport numbers for the two ion species. The transport numbers of a univalent symmetric electrolyte filling the pore space depend on average ion mobilities μ_i and concentrations c_i (e.g., Revil and Glover, 1997)

$$t_p = \frac{\mu_p c_p}{\mu_p c_p + \mu_n c_n}, \quad t_n = \frac{\mu_n c_n}{\mu_p c_p + \mu_n c_n}. \quad (8)$$

Obviously, the effect of different concentrations in the EDL on the total current through the pore is more pronounced in the case of small pore radii, which results in a reduced transport number of the anions particularly in the narrow pore ($t_n < t_p$).

If mobilities are assumed to be the same for both ion species ($\mu_p = \mu_n$, e.g., for KCl), the transport numbers depend on the average concentration only (e.g., Sidorova and Fridrikhsberg, 1978; Buchheim and Irmner, 1979; Titov et al., 2002). Marshall and Madden (1959) considered a 1D system, in which the ion selective behavior of the narrow pore is accounted for by assigning different ion mobilities to wide and narrow pores. Thus, narrow pores become “active” zones, wide pores “passive” zones. Here, differences in ion concentrations are neglected, $c_p = c_n$, and equation 8 reduces to

$$t_p = \frac{\mu_p}{\mu_p + \mu_n}, \quad t_n = \frac{\mu_n}{\mu_p + \mu_n}. \quad (9)$$

Throughout this study, this definition of the transport numbers will be used. However, Bückner and Hördt (2013) recently expanded the originally 1D Marshall-Madden impedance to the case of a 2D system of cylindrical pores. They introduced average transport numbers, which depend on the ion concentration averaged over the cross section of the pore, and thus on pore radii and the properties of

the EDL. Therefore, we keep in mind that the mobility contrasts considered in the present work can be caused not only by true mobility variations, but also by pore radii contrasts and EDL properties.

For small excess ion concentrations (i.e., small electrical fields and currents), Marshall and Madden (1959) solved the 1D correspondent of the above system of differential equations in the frequency domain. The solution is expressed in terms of a complex frequency-dependent impedance (subsequently termed *Marshall-Madden impedance*):

$$Z(\omega) = \frac{\Delta L_1}{\mu_{p1} c_0 F} \times \left[t_{p1} + \frac{B}{A} t_{p2} + \frac{(S_2 - S_1)^2}{\frac{X_1(\omega) S_1}{(t_{p2})^2 t_{p1} \tanh X_1(\omega)} + \frac{A}{B} \frac{X_2(\omega) S_2}{t_{p2} (t_{p1})^2 \tanh X_2(\omega)}} \right]. \quad (10)$$

Here, c_0 is the bulk concentration of the electrolyte in mol/m³ and $\omega = 2\pi f$ the angular frequency in rad/s. The subscript 1 denotes the corresponding quantities of the passive zone and subscript 2 those of the active zone. In addition, the abbreviations

$$X_i(\omega) = \left(\frac{i\omega}{2D_{pi} t_{ni}} \right)^{1/2} \frac{\Delta L_i}{2}, \quad (11)$$

$$A = \frac{\Delta L_1}{\Delta L_2}, \quad (12)$$

$$B = \frac{D_{p1}}{D_{p2}}, \quad (13)$$

and

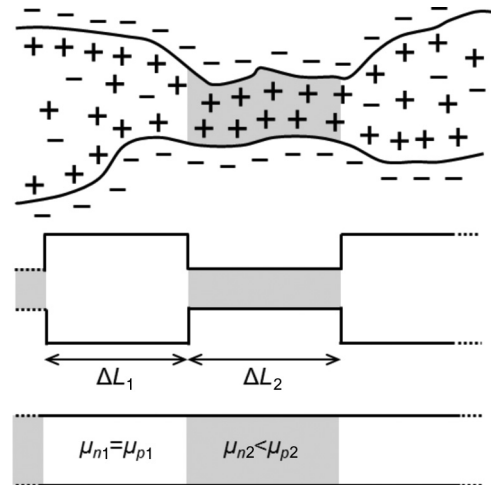


Figure 1. Top: Schematic illustration of the complex pore structure of a natural material with negative surface charge at the mineral-fluid interface and increased cation concentration in the pore fluid. Center: Simplified model consisting in a sequence of two pores with different lengths ΔL_i and cross sections. Bottom: 1D pore space model after Marshall and Madden (1959) consisting in a sequence of two zones of different lengths ΔL_i and ion mobilities μ_{pi} and μ_{ni} .

$$S_i = \frac{t_{ni}}{t_{pi}} \quad (14)$$

have been introduced. The last quantity S_i relates the transport numbers, equation 9, of the respective zones. Note that due to the one-dimensionality of the underlying model, the Marshall-Madden impedance, equation 10, is given as an impedance times unit area (in Ωm^2).

If there was no stronger influence of the EDL on the ion transport numbers in one of the zones (i.e., $S_1 = S_2$), the Marshall-Madden impedance, equation 10, would loose its dependence on frequency and reduce to the simple impedance of two electrolytic conductors in series

$$Z_\infty = \frac{1}{c_0 F} \left[\frac{\Delta L_1}{\mu_{p1} + \mu_{n1}} + \frac{\Delta L_2}{\mu_{p2} + \mu_{n2}} \right]. \quad (15)$$

The same expression can be found in the limiting case of very high frequencies ($\omega \rightarrow \infty$). The case of very low frequencies, i.e., direct current (DC), is treated in the next section.

The Marshall-Madden impedance

Usually, impedance models are formulated in terms of their corresponding DC-limits, polarizabilities (also termed *chargeability*), and time constants. The Marshall-Madden impedance may also be expressed in this form (for a detailed derivation see Appendix A)

$$Z(\omega) = Z_{\text{DC}} \left[1 - \eta_0 \times \left(1 - \frac{\frac{\Delta L_1}{\tau_1} + \frac{\Delta L_2}{\tau_2}}{\frac{\Delta L_1}{\tau_1} \sqrt{i\omega\tau_1} \coth \sqrt{i\omega\tau_1} + \frac{\Delta L_2}{\tau_2} \sqrt{i\omega\tau_2} \coth \sqrt{i\omega\tau_2}} \right) \right], \quad (16)$$

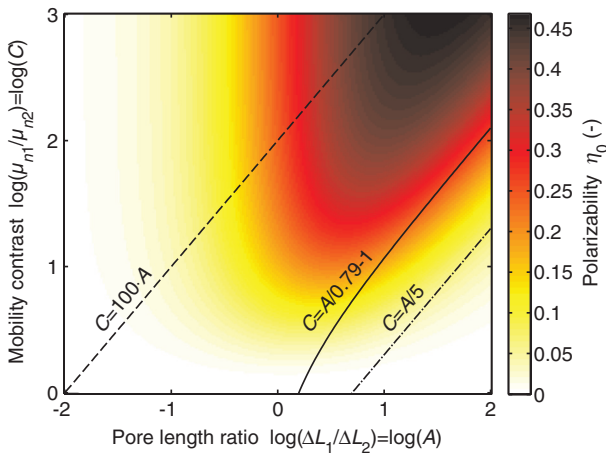


Figure 2. Variation of the polarizability η_0 with the length ratio $A = \Delta L_1/\Delta L_2$ and the mobility contrast $C = \mu_{n1}/\mu_{n2}$. The length of the active zone $\Delta L_1 = 16 \mu\text{m}$ and the mobilities μ_{p1} , μ_{p2} , and μ_{n1} are kept constant, while ΔL_2 and μ_{n2} are varying. Subsequently, the lines will be related to the limits of the LNP regime (dashed line) and the new SNP regime (dash-dot line) as well as the position of the transition between the two regimes (solid line). The basic model parameters are listed in Table 1.

where Z_{DC} is the low frequency limit ($\omega \rightarrow 0$) of the Marshall-Madden impedance, equation 10

$$Z_{\text{DC}} = \frac{k_B T}{e c_0 F} \left[\frac{\Delta L_1}{D_{p1} + D_{n1}} + \frac{\Delta L_2}{D_{p2} + D_{n2}} + \frac{8(t_{n2}t_{p1} - t_{n1}t_{p2})^2}{\frac{\Delta L_1}{\tau_1} + \frac{\Delta L_2}{\tau_2}} \right], \quad (17)$$

and the polarizability is given by

$$\eta_0 = \frac{k_B T}{e c_0 F} \frac{8(t_{n2}t_{p1} - t_{n1}t_{p2})^2}{\frac{\Delta L_1}{\tau_1} + \frac{\Delta L_2}{\tau_2}} \frac{1}{Z_{\text{DC}}}. \quad (18)$$

Here, all mobilities have been expressed in terms of the corresponding diffusion coefficients using the Nernst-Einstein relation (equation 6). The dependence on frequency, formerly represented by X_i , is now expressed in terms of time constants

$$\tau_i = \frac{\Delta L_i^2}{8D_{pi}t_{ni}} = \frac{\Delta L_i^2(D_{pi} + D_{ni})}{8D_{pi}D_{ni}}. \quad (19)$$

In this way, two new time constants, one for each zone, are defined. Note that, throughout this study, the term *time constant* refers to the physical quantities τ_i that directly appear in the impedance models and have the dimension of time. The term *characteristic time* describes the dominant time scale of the relaxation process and is calculated as the inverse of the spectral position of the phase maximum $\tau_c = 1/\omega_c$. Some authors use the term *relaxation time* for this quantity. In accordance with previous studies (e.g., Titov et al., 2002; Blaschek and Hördt, 2009; Volkman and Klitzsch, 2010), both time constants (equation 19) show a quadratic dependence on the respective pore length and are inversely proportional to a diffusion coefficient. The impedance is invariant with regard to a permutation of the pore indices (1, 2). This symmetry is broken, as soon as limiting cases are considered. Examples are the SNP model by Titov et al. (2002) (only valid for $\Delta L_1 \gg \Delta L_2$) and the new models proposed further ahead in this study.

The polarizability reflects the magnitude of the polarization effect. It is worthwhile to review some aspects of the polarizability's dependence on the model parameters by studying its variation with the pore length ratio $A = \Delta L_1/\Delta L_2$ and the ratio of the anion mobilities in the two pores $C = \mu_{n1}/\mu_{n2}$. Note that C is the anionic analog of B (see equation 13) and will subsequently be termed *mobility contrast*. Figure 2 shows that high polarizabilities are caused by reduced mobilities in the shorter pore, i.e., by short active zones (narrow passages, $A \gg 1$). However, high pore length ratios A only lead to high polarizabilities, if the mobility contrast is increased too. These findings agree with those derived from numerical simulations (cf., Blaschek and Hördt, 2009). The significance of the lines and the corresponding equations specified in the figure will be explained further below.

Instead of the polarizability, the full frequency spectrum of the Marshall-Madden impedance can be analyzed. Figure 3 shows huge phase values for mobility contrasts $C > 1$, where the short pore is the active zone. Maximum phase values increase with increasing mobility contrast and saturate at about $C \approx 10^3$. In the range $C < 1$, phases are much smaller and phase maxima are shifted by approximately one order of magnitude to higher frequencies.

To enhance the phases and facilitate analytic derivations in this study, we will use high mobility contrasts C between 100 and 500 (see Table 1). Note that maximum phases of up to -200 mrad as reached for the chosen mobility contrasts are much higher than those found experimentally for samples without metallic constituents. As mentioned above, we understand the low anion mobility in the active zone as a parameterization of the average anion transport number (see Bucker and Hördt, 2013). Therefore, high mobility contrasts may be caused by high contrasts in pore radius and ion concentration in addition to true mobility contrasts.

The variation of the phase spectra with the pore lengths, as illustrated in Figure 4, reveals a basic property of the Marshall-Madden impedance. The spectral position of the phase maxima varies with the lengths of both pores. However, the dependence is restricted to pore lengths smaller than a limit value. In Figure 4a, the phase maxima vary with the lengths of the wide pore as long as $\Delta L_1 < 10^{-3}$ m, and in Figure 4b with the lengths of the narrow pore as long as $\Delta L_2 < 10^{-7}$ m.

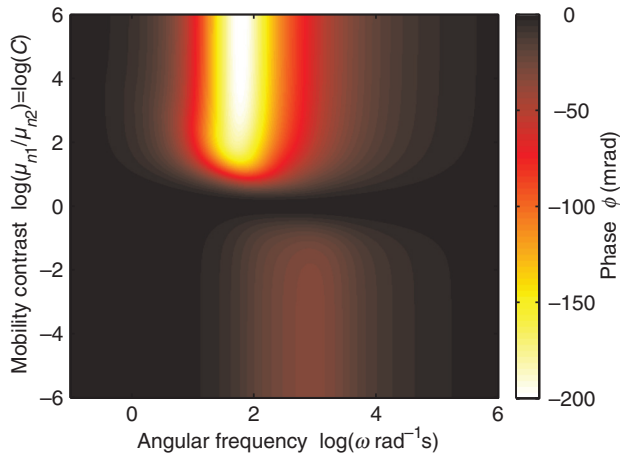


Figure 3. Variation of the phase ϕ of the Marshall-Madden impedance with the frequency ω and the mobility contrast $C = \mu_{n1}/\mu_{n2}$. The mobilities μ_{p1} , μ_{p2} , and μ_{n1} are kept constant, while μ_{n2} is varying. Here, $\Delta L_1 = 16 \mu\text{m}$ and $\Delta L_2 = 4 \mu\text{m}$.

Table 1. Basic model properties for Figures 2–11. Any property that differs from its value indicated in this table is specified in the caption to the corresponding figure. Ionic mobilities are close to the one of Na^+ in dilute solution at 25°C ($\mu_{\text{Na}^+} = 5.19 \cdot 10^{-8} \text{ m}^2/(\text{Vs})$). The reduction of the anion mobility in the active zone by 1/100 was selected to yield a significant mobility contrast.

Property	Symbol	Value
Absolute temperature	T	291 K
Concentration bulk electrolyte	c_0	1 mol/m
Anion mobility, passive zone	μ_{n1}	$5 \cdot 10^{-8} \text{ m}^2/(\text{Vs})$
Anion mobility, active zone	μ_{n2}	$5 \cdot 10^{-10} \text{ m}^2/(\text{Vs})$
Cation mobility, passive zone	μ_{p1}	$5 \cdot 10^{-8} \text{ m}^2/(\text{Vs})$
Cation mobility, active zone	μ_{p2}	$5 \cdot 10^{-8} \text{ m}^2/(\text{Vs})$

For lengths lesser and greater than the limit value, the positions of the phase maxima form straight lines. The slopes of lines g and h, both ≈ -0.5 , can be used to derive preliminary relations between pore lengths and characteristic times. Via $\tau_c = 1/\omega_c$, the spectral position of the phase maximum ω_c is related to the characteristic time τ_c . For line g, we find a dependence on the length of the active zone $\tau_c \sim \Delta L_1^2$. This relation corresponds to the one reported by Blaschek and Hördt (2009) and Volkmann and Klitzsch (2010). On the other hand, line h leads to a quadratic variation of the characteristic time with the length of the narrow pore $\tau_c \sim \Delta L_2^2$ as found earlier by Titov et al. (2002). These preliminary results suggest that both behaviors are included in the Marshall-Madden impedance and can be derived as limiting cases.

For a more detailed analysis of the relation between the characteristic time and the two pore lengths, the characteristic time is computed based on the spectra of every pair of pore lengths. Although in this representation all information on magnitude and shape of the phase spectra is lost, Figure 5 allows a clear distinction

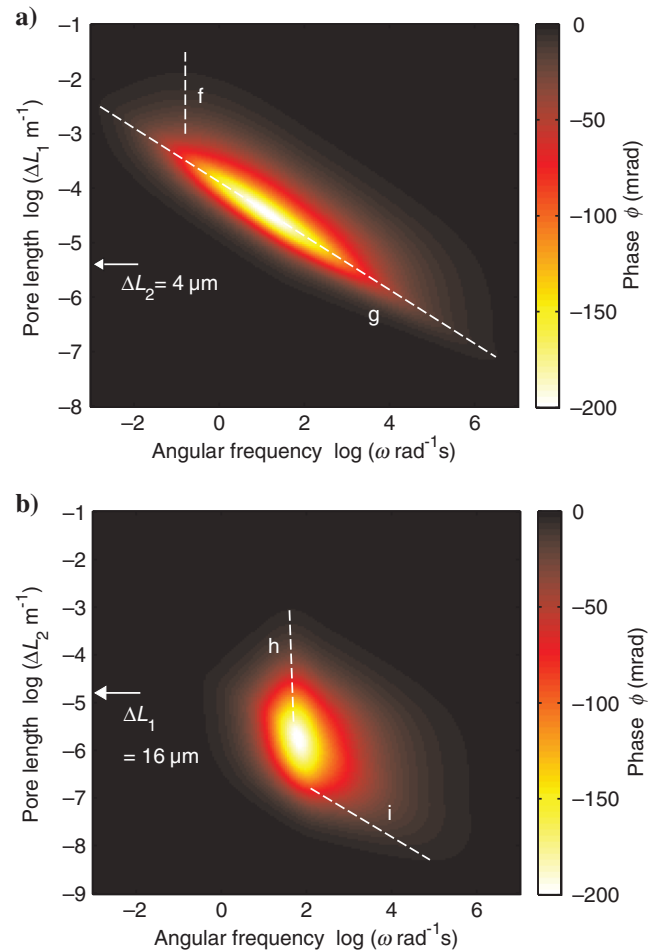


Figure 4. (a) Variation of the phase ϕ of the Marshall-Madden impedance with the frequency and the length ΔL_1 of the passive zone. The length of the active zone $\Delta L_2 = 4 \mu\text{m}$ is kept constant. (b) Variation of the phase ϕ with the frequency and the length ΔL_2 of the active zone. Here, the length of the passive zone $\Delta L_1 = 16 \mu\text{m}$ is kept constant. Lines “g” and “h” indicate a trend of the spectral position of the phase peak varying as a function of the respective pore length. For greater lengths, this trend disappears as indicated by lines “f” and “i”.

between two regimes: For $A \leq C/100$ (long active zone and/or large mobility contrast), the characteristic times only vary with the length ΔL_1 of the passive zone. Again, in this regime the relation $\tau_c \sim \Delta L_1^2$ holds. For $A \geq 5 \cdot C$ (long passive zone), the characteristic times solely depend on the length ΔL_2 of the active zone. As expected, this dependence can be expressed as $\tau_c \sim \Delta L_2^2$. In the range between these two regimes, the characteristic time varies with both lengths simultaneously. The formulation of the criteria in terms of the mobility contrast C ($A \leq C/100$, $A \geq 5 \cdot C$) is illustrated here for $C = 100$, but was verified for different values of C between 1 and 1000.

TWO NEW IMPEDANCE MODELS

Although the preceding analysis showed that the characteristic time versus pore length relation of the Marshall-Madden impedance is composed of two very simple functional relationships, no sufficiently simple analytical expression could be derived for the whole range of pore lengths. However, simple analytical expressions are made possible by the derivation of two new approximate impedance models for the two limiting cases $A \leq C/100$ and $A \gg 5 \cdot C$.

A long narrow pore model

Based on the reformulated Marshall-Madden impedance given in equation 16, an approximate impedance model for the regime $A \leq C/100$ and approximately equal mobilities $\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$ is derived (see Appendix A). Because this regime is characterized by relatively long narrow pores, the corresponding model shall be termed the *long narrow pore (LNP) model*:

$$Z_{\text{LNP}}(\omega) = Z_{\text{DC}} \left[1 - \eta_0 \left(1 - \frac{\tanh \sqrt{i\omega\tau_1}}{\sqrt{i\omega\tau_1}} \right) \right]. \quad (20)$$

The DC-limit, polarizability, and time constant are the same as for the Marshall-Madden impedance and given by equations 17, 18, and 19, respectively.

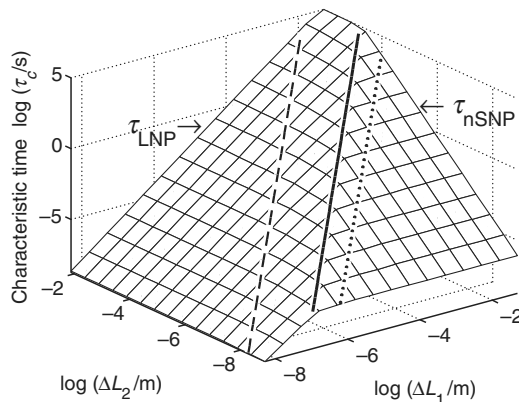


Figure 5. Variation of the characteristic time τ_c with simultaneously varying pore lengths ΔL_1 and ΔL_2 . The mobilities are chosen to provide a high-mobility contrast of $C = \mu_{n1}/\mu_{n2} = 100$ and high polarizabilities. Note the clear separation of the two regimes $A < C/100$ (delimited by the black dashed line) and $A > C/5$ (black dotted line). The black solid line ($A = 0.79 \cdot (C + 1)$) illustrates the approximate position of the transition between the two regimes.

Figure 6 illustrates the good agreement of the new LNP model with the Marshall-Madden impedance. Under the given condition $A \leq C/100$, a quantitative analysis of the deviations confirmed a small relative error $< 3\%$ for the magnitude of the impedance (not shown here for brevity). Note that, in this limiting case, the phase has a clearly asymmetrical shape. The low-frequency branch of the phase dispersion increases proportionally to ω while the high-frequency branch decreases at $-1/\sqrt{\omega}$. In terms of empirical models, asymmetric dispersions are often fitted by Cole-Davidson models. Effectively, the impedance spectrum matches a generalized Cole-Cole model, equation 1, if the time constant is $\tau_{\text{GCC}} = \tau_1$ and the exponents are $a = 0.5$ and $c = 1$ (Cole-Davidson model).

Figure 7 shows that the time constant of the impedance model does not necessarily coincide with its characteristic time. The variation of the characteristic times with the pore length ΔL_1 in Figure 7 was computed on the basis of spectra for each pore length. For each of the respective phase spectra, the characteristic angular frequency was determined and the characteristic time calculated as $\tau_c = 1/\omega_c$.

Due to the good agreement between the corresponding phase spectra, the characteristic times of the Marshall-Madden impedance

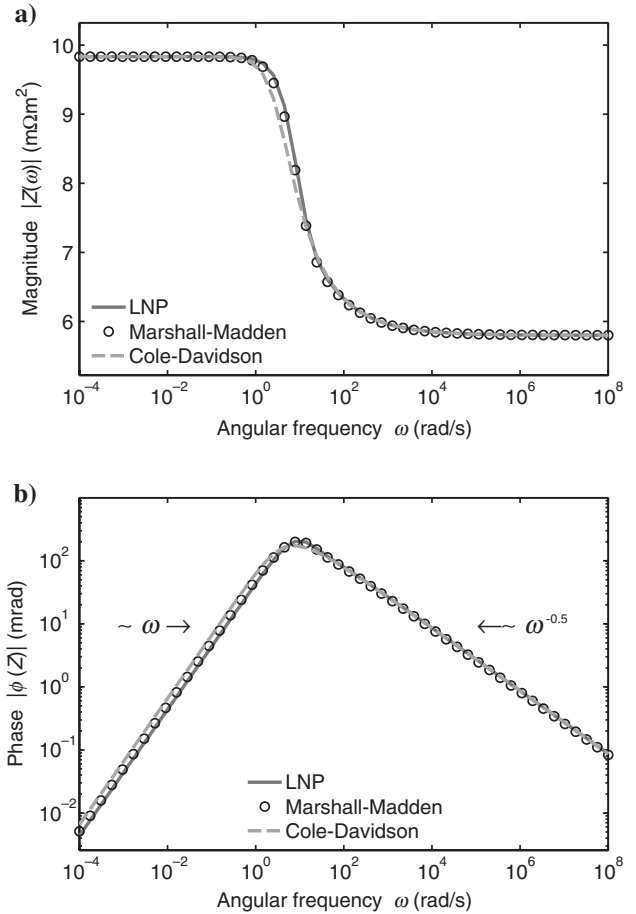


Figure 6. (a) Magnitude and (b) phase spectra of the Marshall-Madden impedance (circles), the LNP model (black solid line), and the Cole-Davidson model (gray dashed line) according to equation 1, where $c = 1$, $a = 0.5$, and $\tau_{\text{GCC}} = \tau_1$. The pore lengths $\Delta L_1 = 40 \mu\text{m}$ and $\Delta L_2 = 8 \mu\text{m}$ as well as the anion mobility in the active zone $\mu_{n2} = 1 \cdot 10^{-10} \text{ m}^2/(\text{Vs})$ are chosen to match the conditions $A \leq C/100$ and $C \gg 1$.

(see Figure 7) are recovered quite well by those of the LNP model. For pore lengths $\Delta L_1 > 10^{-3}$ m, the assumption $A \leq C/100$ is violated and the transition to the SNP model (no variation with ΔL_1 , see next section) explains the increasing differences. We also see in Figure 7 that the actual characteristic time of the LNP model, equation 20, is smaller than the time constant τ_1 . As shown in Appendix A, this can be explained theoretically. The corresponding factor between the actual characteristic time of the model τ_{LNP} and the time constant τ_1 is (<1)

$$\tau_{\text{LNP}} \approx [0.0222 + (0.1374 - 0.1245\eta_0)^{1/2}] \tau_1. \quad (21)$$

This analytically derived characteristic time is proportional to $\tau_1 = \Delta L_1^2 / (8D_{p1}t_{n1})$ for a given polarizability, which explains the observed quadratic dependence of the characteristic time in the LNP regime on the length of the wide (passive) pore.

The characteristic time of the LNP model, equation 21, may be compared to the empirical relationship reported by Blaschek and Hördt (2009), whose corresponding equation may be expressed as

$$\tau_c = 0.0998 \frac{\Delta L_1^2}{D}, \quad (22)$$

based on various numerical simulations. Here, D denotes the diffusion coefficient in the active zone ($D = D_{p1} = D_{n1}$) and τ_c is the characteristic time. A similar expression may be derived for our analytic LNP model. To this end, we consider the limiting case of very small polarizabilities $\eta_0 \ll 1$ in which the characteristic time, equation 21, reduces to $\tau_{\text{LNP}} \approx 0.4 \cdot \tau_1$. We get

$$\tau_{\text{LNP}} \approx 0.4 \cdot \tau_1 = 0.4 \cdot \frac{\Delta L_1^2}{8D_{p1}t_{n1}} \approx 0.1 \frac{\Delta L_1^2}{D}, \quad (23)$$

where in the last step $D = D_{p1} = D_{n1}$ was used to yield $t_{n1} = 0.5$. Both relations, equations 22 and 23, match very well and thus the characteristic time of the LNP model, equation 21, represents a theoretical explanation of the numerical findings by Blaschek and Hördt (2009).

A new short narrow pore model

The regime $A \geq 5 \cdot C$ is characterized by relatively SNP. The existing SNP model by Titov et al. (2002) offers a good approximation of the impedance for very pronounced length ratios $A \gg 1$. However, its applicability is limited to small mobility contrasts $C \approx 1$. In this section, a new short narrow pore (nSNP) model is presented, which overcomes this limitation. Instead, the condition for the length ratio is formulated in terms of the mobility contrast, as our previous discussion suggests $A \gg 5 \cdot C$.

An approximate SNP impedance model, which is able to predict the relation between characteristic time and pore lengths, must hold at least in the vicinity of the phase maximum. For frequencies ω close to the phase maximum at ω_c the condition $A \gg 5 \cdot C$ with the additional assumptions of approximately equal mobilities $\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$ and reduced anion mobilities in the active zone $\mu_{n1} > \mu_{n2}$ imply $\omega\tau_1 \gg 1$ and $\omega\tau_2 < 1$ (see Appendix A). Under these five assumptions, based on the reformulated Marshall-Madden impedance, equation 16, an nSNP can be derived (Appendix A) given by

$$Z_{\text{nSNP}}(\omega) = Z_{\text{DC}} \left[1 - \eta_0 \left(1 - \frac{1}{1 + \sqrt{i\omega\tau'}} \right) \right]. \quad (24)$$

Here again, the DC-limit and the polarizability are given by equations 17 and 18. The modified time constant

$$\tau' = \tau_2 \frac{D_{p1}t_{n1}}{D_{p2}t_{n2}} = \frac{\Delta L_2^2}{8D_{p2}t_{n2}} \frac{D_{p1}t_{n1}}{D_{p2}t_{n2}} \quad (25)$$

depends on the length of the active zone and on the ion mobilities in both zones. Note that the nSNP model is equivalent to a Cole-Cole model with the frequency exponent $c = 0.5$.

Figure 8a shows a good agreement between the magnitude spectra of our nSNP model and the Marshall-Madden impedance. A more extensive analysis revealed a relative error $<1\%$, as long as the condition $A \gg 5C$ is fulfilled. Due to the unfavorable combination of pore length ratio and mobility contrast ($A \geq 5 \cdot C$, lower right corner of Figure 2), spectral changes in the magnitude (i.e., polarizabilities) are much smaller than in the LNP regime. Note that the chosen scaling of the ordinate exaggerates existing deviations between the models.

The phase dispersion of the nSNP model in Figure 8b shows a symmetric behavior, which is typical of the Cole-Cole model. The low-frequency branch increases proportionally to $\sqrt{\omega}$, whereas the high-frequency branch decreases at a rate of $-\sqrt{\omega}$. The phases of the nSNP model and the Marshall-Madden impedance match very well, except for the low-frequency range, where the model assumption $\omega\tau_1 \gg 1$ is violated and the phase of the Marshall-Madden impedance shows an ω -like behavior. The poor agreement with the original SNP model by Titov et al. (2002) is due to the violation of the additional condition $C \approx 1$.

The deviations in the low-frequency range, the phase maximum, and therefore the characteristic frequency, are reproduced very well. As long as the model assumptions are fulfilled, this results in the good agreement between the characteristic times of both models in Figure 9. In the upper range of pore lengths ($\Delta L_1 > 2 \cdot 10^{-6}$ m),

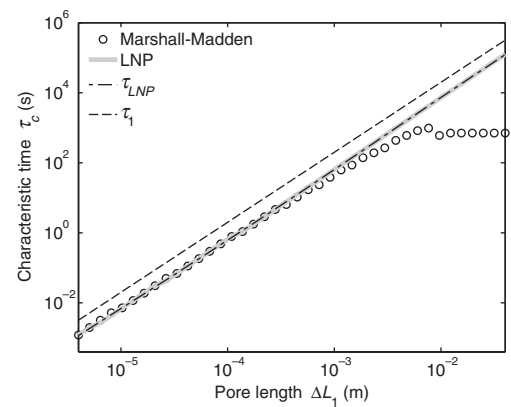


Figure 7. Variation of the characteristic times τ_c with the length ΔL_1 of the active zone. The characteristic times of the Marshall-Madden impedance (circles) and the LNP model (gray solid line) were computed based on the characteristic frequencies ω_c of spectra for each pore length ΔL_1 . The characteristic time τ_{LNP} (black dashed-dot line) was calculated according to equation 21. The time constant τ_1 (black dashed line) is given by equation 19. The pore length $\Delta L_2 = 8 \mu\text{m}$ and the mobility $\mu_{n2} = 1 \cdot 10^{-10} \text{ m}^2/(\text{Vs})$ are chosen to match the conditions $A \leq C/100$ and $C \gg 1$.

the characteristic times of the Marshall-Madden impedance tend toward a constant value (transition to LNP regime). This trend is not followed by the characteristic time of the nSNP model. As shown in Appendix A, the characteristic time can be expressed in terms of the modified time constant τ' and the polarizability

$$\tau_{nSNP} = (1 - \eta_0)\tau'. \quad (26)$$

For small polarizabilities $\eta_0 \ll 1$, the characteristic time τ_{nSNP} can be approximated by the modified time constant τ' .

Titov's short narrow pore model

Although Titov et al. (2002) developed the SNP model in the time domain, their approach is based on the membrane polarization theory by Marshall and Madden (1959). Therefore, it is possible to derive the SNP model starting from the frequency-domain Marshall-Madden impedance, equation 16. Adding the assumption $\omega\tau_1 \gg 1$, which is partly implied by their conditions $A \gg 1$ and $C \approx 1$, we obtain (Appendix A)

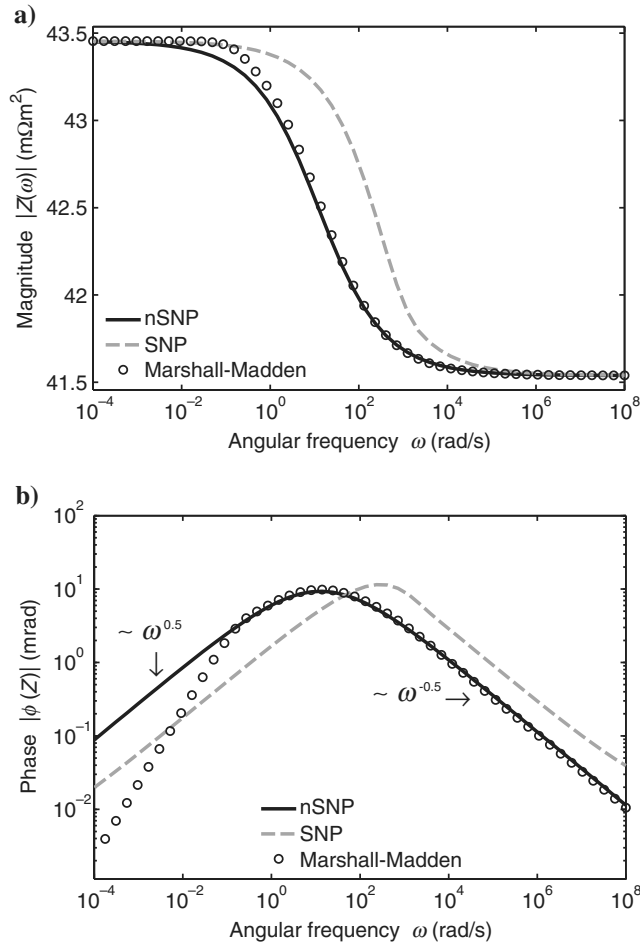


Figure 8. (a) Magnitude and (b) phase spectra of the Marshall-Madden impedance (circles), Titov's SNP model (gray dashed line), and the nSNP model (black solid line). The pore lengths $\Delta L_1 = 400 \mu\text{m}$ and $\Delta L_2 = 0.4 \mu\text{m}$ as well as the anion mobility in the active zone $\mu_{n2} = 1 \cdot 10^{-10} \text{ m}^2/(\text{Vs})$ are chosen to match the conditions $A \gg 5C$ and $C \gg 1$.

$$Z_{\text{SNP}}(\omega) = Z_{\text{DC}} \left[1 - \eta_0 \left(1 - \frac{1 - \exp(-2(i\omega\tau_2)^{1/2})}{2(i\omega\tau_2)^{1/2}} \right) \right], \quad (27)$$

where the time constant is given by

$$\tau_2 \approx \frac{\Delta L_2^2}{4D_{n2}}. \quad (28)$$

This is essentially the same impedance model as the one obtained by Titov et al. (2002) after the transformation to the frequency domain. The main difference between the two models consists of the expressions for the DC-limits of the impedances and the respective polarizabilities. In our approach, both quantities are given by the 1D Marshall-Madden impedance and expressed in terms of pore lengths and the transport numbers of both pores (see equations 17 and 18). The corresponding quantities in the original model by Titov et al. (2002) are given in terms of the same parameters plus the pore cross sections and an efficiency, which accounts for the different anion and cation concentrations in the EDL. That is why for $C \approx 1$ Titov et al. (2002) obtained reasonable polarizabilities (0.5%–2%) while the Marshall-Madden impedance predicts values close to zero (see Figure 2).

An additional minor difference is the choice of the diffusion coefficient for the calculation of the time constant. Although in the original model it is expressed in terms of the diffusion coefficient of the bulk electrolyte, our approach yields a dependence on the anionic diffusion coefficient in the narrow pore. For the case of small mobility contrasts (where the SNP model matches the corresponding limiting case of the Marshall-Madden impedance), this detail does not make a numeric difference, but is essential for the understanding of the underlying process.

The poor agreement between the respective magnitude and phase spectra of the SNP and the Marshall-Madden model in Figure 8 improves when the assumption $C \approx 1$ of the SNP model is fulfilled (not shown here).

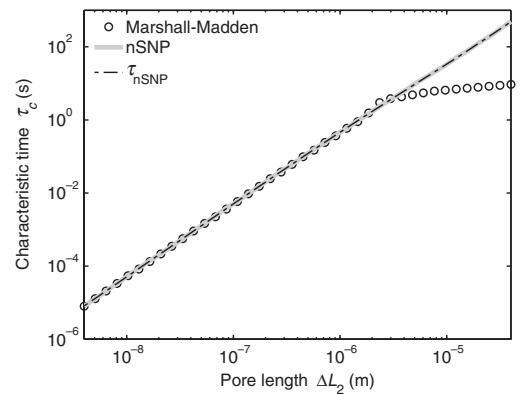


Figure 9. Variation of the characteristic times τ_c with the length ΔL_2 of the active zone. The characteristic times of the Marshall-Madden impedance (circles) and the nSNP model (gray solid line) were computed based on the characteristic frequencies ω_c of spectra for each pore length ΔL_2 . The characteristic time τ_{nSNP} (black dashed line) was calculated according to equation 21. The pore length $\Delta L_1 = 400 \mu\text{m}$ and the anion mobility in the active zone $\mu_{n2} = 1 \cdot 10^{-10} \text{ m}^2/(\text{Vs})$ are chosen to match the conditions $A \gg 5C$ and $C \gg 1$.

For small polarizabilities ($\eta_0 \ll 1$) — which are expected for the low mobility contrasts required to fulfill the SNP condition — the dependence of the characteristic time of the SNP impedance on the time constant τ_2 can be derived analytically (Appendix A)

$$\tau_{\text{SNP}} \approx 1.12 \cdot \tau_2. \quad (29)$$

The misfit between τ_{SNP} and the characteristic time of the SNP impedance in Figure 10 can only be reduced by decreasing the mobility contrast even more. The characteristic time of the nSNP model τ_{nSNP} , on the other hand, shows a much better agreement with the Marshall-Madden impedance.

In a mathematically strict sense, in particular with respect to the predicted characteristic time, the validity of the original SNP model is restricted to a narrow range of small mobility contrasts. Our nSNP model shares the main features of the SNP model at low mobilities, but overcomes its limitations and describes the impedances for high and low mobility contrasts at the same time.

IMPLICATIONS OF THE TWO NEW IMPEDANCE MODELS

Two characteristic time regimes

In other contexts, quantities of the type $\sim x^2/4D$, where x is a length and D a diffusion coefficient, are known as diffusion times. The diffusion time represents the time scale, on which a concentration distortion propagates along a profile of the length x . In the case of membrane polarization, the diffusion time — our characteristic time — controls the buildup and relaxation of excess and defect concentrations at the zone boundaries. For both SNP models, the dependence of the characteristic time was found to be basically $\sim \Delta L_2^2/D_{n2}$. In this case, the relaxation process seems to be dominated by diffusion via the narrow pore. For the LNP model, we

arrived at the relation $\sim \Delta L_1^2/D_{n1}$, which indicates the predominance of diffusion relaxation via the wide pore.

To understand the existence of these two characteristic time regimes, we revisit the variation of the characteristic time with both pore lengths. The 3D representation of the characteristic time in Figure 5 is dominated by two planar surfaces. Within the parameter range of each regime, these surfaces are given by the two characteristic times, respectively. The geometrical shape of the entire function can be reproduced by intersecting the two planes $\tau_{\text{LNP}}(\Delta L_1, \Delta L_2)$ and $\tau_{\text{nSNP}}(\Delta L_1, \Delta L_2)$ and removing the half-planes representing characteristic times that are bigger than the ones on the other plane. Physically speaking, in each case, the shorter characteristic time dominates the relaxation process and excess (defect) concentrations at the zone boundaries are mainly built up and balanced via the diffusion path with the shorter diffusion time.

In a more quantitative manner, this results in a criterion for the discrimination of the two regimes. The transition occurs approximately where the two planes intersect. Equating the two characteristic times (equations 21 and 26) gives

$$A = \frac{\Delta L_1}{\Delta L_2} = \frac{1}{\sqrt{0.4}} \frac{D_{p1} t_{n1}}{D_{p2} t_{n2}} \approx 0.79 \cdot (C + 1). \quad (30)$$

Here, both characteristic times have been taken in their limits for small polarizabilities $\tau_{\text{LNP}} \approx 0.4\tau_1$ and $\tau_{\text{nSNP}} \approx \tau'$ and the ratio of the diffusion coefficients has been approximated by $D_{p1}/D_{p2} \approx 1$. The ratio t_{n1}/t_{n2} can be approximated by $(C + 1)/2$, following from $\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$, which yields $t_{n1} \approx 0.5$ and $t_{n2} \approx 1/(C + 1)$. Figure 5 shows a good agreement between the corresponding line $A = 0.79 \cdot (C + 1)$ and the real position of the transition between the two regimes.

The existence of this criterion affirms the dependence of the model assumptions ($A \leq C/100$ and $A \geq 5 \cdot C$) on the mobility contrast C , which had only been suggested from empirical studies so far.

Transition between the two relaxation time regimes

The impedance spectra of the LNP model and the nSNP model do not only differ with respect to their characteristic times, but also show clearly distinct characteristic shapes. Because it represents a special case of the Cole-Cole model, the nSNP model inherits the typical symmetrical behavior. The LNP model, in contrast, has clearly asymmetrical spectra similar to those of a Cole-Davidson model. In addition, the LNP model shows a narrower phase peak than the nSNP model, which goes along with a steeper decrease in the magnitude spectrum at the characteristic frequency. Beside the existing differences, the high-frequency behavior of the phase spectra of both models is controlled by the same $1/\sqrt{\omega}$ -decrease. Most of the behavior of the Marshall-Madden impedance can be described and understood based on these two fundamental constituents.

The following comparison with the phase spectra of the Marshall-Madden impedance explains some of the mentioned differences and agreements between the two models. In Figure 11, phase spectra for different lengths of the wide pore are plotted simultaneously, illustrating the evolution of the Marshall-Madden impedance from the LNP regime (small ΔL_1) to the SNP regime (great ΔL_1). Beginning with the spectra corresponding to long wide pores (great ΔL_1), the symmetrical shape of the SNP phases can be

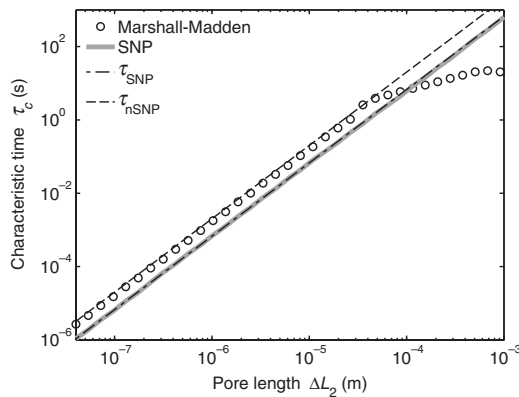


Figure 10. Variation of the characteristic times τ_c with the length ΔL_2 of the active zone. The characteristic times of the Marshall-Madden impedance (circles) and the SNP model (gray solid line) were computed based on the characteristic frequencies ω_c of spectra for each pore length ΔL_2 . The characteristic time τ_{SNP} (black dashed-dot line) was calculated according to equation 29. The characteristic time of the nSNP model τ_{nSNP} (black dashed line) is given by equation 26. Even though the pore length $\Delta L_1 = 400 \mu\text{m}$ and the anion mobility in the active zone $\mu_{n2} = 1 \cdot 10^{-8} \text{ m}^2/(V_s)$ are chosen to match the conditions $A \gg 1$ and $C \approx 1$, τ_{nSNP} matches the characteristic time of the Marshall-Madden impedance slightly better than τ_{SNP} .

recognized. Here, the SNP conditions are fulfilled, and while the phase maximum increases with decreasing lengths of the wide pore, it remains at its spectral position. The sharp bend at low frequencies, which is not included in the SNP impedance and was attributed to the violation of the assumption of relatively high frequencies ($\omega\tau_{n\text{SNP}} \ll 1$) earlier, approaches the phase maximum. It is obvious that, at low frequencies, the whole set of possible phase spectra is confined by the characteristic ω -dependence of the LNP phase. The transition between the two regimes coincides with the length of the wide pore, for which the bend reaches the phase maximum and the phase peak begins to wander toward higher frequencies ($\Delta L_1 = 0.1$ mm). The resulting envelope for the whole set of possible phase spectra is confined by $\sim\omega$ at low frequencies and $\sim 1/\sqrt{\omega}$ at high frequencies.

Strictly speaking, the two new impedance models are limited to their respective regimes and to defined frequencies that extend from a proximity of the phase maximum to high frequencies.

Practical significance

A summary of the membrane polarization impedances based on the model by Marshall and Madden (1959), including their respective characteristic times, is given in Table 2. Beside providing a basic understanding of the fundamental processes, they may also serve as generic models to calculate the impedances of more realistic pore geometries. The answer to the question whether any of the

impedances is more relevant for practical applications is beyond the scope of this paper. Intuitively, the two SNP models might seem more suitable, as the narrow passages are generally imagined to be relatively short. However, realistic pore geometries can be very complex, and the simulation based on fundamental models is not trivial. Also note that, up to now, only one mobility has been changed during the derivation of the relaxation models. More complication might arise if an extension of the model family might be necessary after having dropped the assumption $D_{p1} = D_{p2} = D_{n1}$. At this point, both models are relevant.

With respect to the interpretation of spectra obtained from real materials, the two new impedance models in the first line offer a strong physical link between the empirical models of the Cole-Cole family, and in this sense legitimize their application. In addition, they facilitate the discussion of the characteristic shape of experimentally obtained spectra. The widely used quadratic dependence of the characteristic time on the length scale of the pore space has been confirmed.

In real materials, simple relaxation or characteristic time relations are not always able to explain the experimental observations. Kruschwitz et al. (2010) and Titov et al. (2010), for instance, recently reported the presence of long characteristic times in samples with relatively small pore throat diameter (<7 μm), which do not obey any of the theoretically predicted relationships. Basically, two mechanisms were speculated to cause these unexpected observations. Ionic mobility in the EDL is reduced in comparison to the bulk electrolyte. In materials with small pore sizes and a high specific surface area, the relatively slow surface diffusion might predominate and increase the observed characteristic time considerably. The second mechanism is based on the idea that narrow pores could form a connected pathway, which acts as one long membrane zone. Both mechanisms would result in an increased characteristic time.

Especially for the second mechanism, the consequences of the present study are relevant. One of our main results is that the relaxation process is dominated by the shorter characteristic time. The observation of long characteristic times can therefore only be explained by assuming that even the shortest characteristic time is relatively long. The idea of interconnected narrow passages, which form one long active zone, is reminiscent of the LNP model. However, in the LNP model, characteristic times are determined by the length of the wide pore. To generate long characteristic times, as they were found experimentally, the wide pore would have to be long as well. Therefore, assuming the LNP model, one would arrive at a pore space with approximately equal lengths of the wide and the narrow pore. An interpretation based on the SNP model would result in

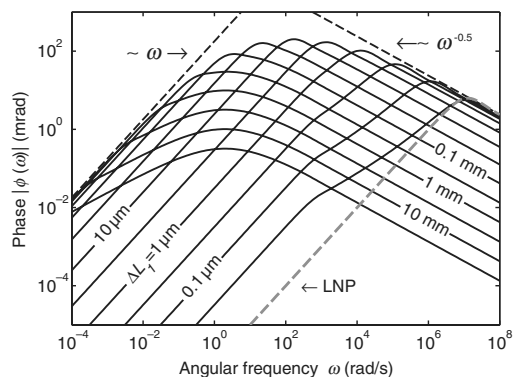


Figure 11. Variation of the phase spectra with the length ΔL_1 of the wide pore. The length of the narrow pore $\Delta L_2 = 1$ μm is kept constant. The gray dashed line represents the phase response of the LNP model for the smallest value of ΔL_1 . The two black dashed lines mark the low- and high-frequency envelopes of the whole set of phase spectra.

Table 2. Summary of the assumptions made for the derivation of the impedance models and the corresponding characteristic times. With respect to the length ratio, the mobility contrast, and the remaining mobilities, no assumptions are made for the Marshall-Madden impedance, which is the most general model.

Model	Length ratio	Mobility contrast	Remaining mobilities	Characteristic time
Marshall-Madden	—	—	—	—
LNP (this work)	$A < C/100$	—	$\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$	$\tau_{\text{LNP}} \approx 0.4 \cdot \frac{\Delta L_1^2}{8D_{p1}t_{n1}}$
SNP (Titov)	$A \gg 1$	$C \approx 1$	$\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$	$\tau_{\text{SNP}} \approx 1.12 \cdot \frac{\Delta L_2^2}{4D_{p2}}$
nSNP (this work)	$A \gg 5C$	—	$\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$	$\tau_{\text{nSNP}} \approx \frac{\Delta L_2^2}{8D_{p2}t_{n2}} \frac{D_{p1}t_{n1}}{D_{p2}t_{n2}}$

even longer wide pores or in interconnected wide pores, which are separated by short narrow passages. Any of the two models would require interconnected wide pores as well as interconnected narrow passages.

Another interesting experimental result was recently reported by Weller et al. (2010a), who observed a negative correlation between measured permeability and characteristic time (their mean relaxation time) for a set of sandstone samples. They explained the increase of the characteristic time by longer pathways in narrow pores. However, it seems unlikely that this explanation is consistent with this membrane polarization model. The present study shows that, in this model, the characteristic time is usually dominated by the shorter time constant, which in presence of LNP — as assumed by Weller et al. (2010a) — would be the one of the wider pores. An explanation might have to consider the expanded Marshall-Madden impedance developed in the work of Bückner and Hördt (2013). Their studies are focused on the effect of pore radii, and they did not find a negative correlation between characteristic time and pore radii, which might explain the observation of Weller et al. (2010a). However, true mobility variations caused by concentration dependence in the wide and narrow pores might also be important. A decrease of mobilities in narrower pores due to an increase of concentration might cause increasing time constants in both types of pores. This effect was not implemented in the model by Bückner and Hördt (2013), and thus it remains unclear at this stage whether, and under which conditions this effect is significant.

EQUIVALENT CIRCUIT REPRESENTATION OF THE MARSHALL-MADDEN IMPEDANCE

Equivalent circuits are often used for the illustration and interpretation of impedance models (e.g., Pelton et al., 1978; Wong, 1979). Writing the Marshall-Madden impedance, equation 16, as

$$Z(\omega) = Z_{\infty} + [1/Z_{W1}(\omega) + 1/Z_{W2}(\omega)]^{-1}, \quad (31)$$

reveals that it can be represented by an equivalent circuit with the high-frequency limit Z_{∞} (equation 15) in series with two impedances

$$Z_{Wi}(\omega) = Z_{Di} \frac{\tanh \sqrt{i\omega\tau_i}}{\sqrt{i\omega\tau_i}} \quad (32)$$

in parallel. These impedances have the form of finite-length Warburg impedances (e.g., Franceschetti et al., 1991) that are characteristic for diffusion processes in cases where the diffusion length $L_d \propto \sqrt{D/\omega}$ (i.e., distance concentrations propagate due to diffusion in time $\tau = 1/\omega$) is equal to or greater than the length of the region available for diffusion. The magnitudes of the finite-length Warburg impedances,

$$Z_{Di} = \frac{k_B T}{ec_0 F} 8(t_{n2} t_{p1} - t_{n1} t_{p2})^2 \frac{\tau_i}{\Delta L_i}, \quad (33)$$

are the maximum diffusion-related resistance of the corresponding pores. Note that this diffusion resistance does not only depend on the length and the diffusion coefficients of the pore i (see definition of time constant, equation 19) but also on the transport numbers of both pores.

CONCLUSIONS

The Marshall-Madden impedance as a model for membrane polarization bears a whole family of impedance models. We cast it into the classical form involving polarizability, and find that two time constants are required to describe the spectral behavior. To identify the dominating time scales and corresponding length scales, we propose approximate models, which allow for analytical expressions describing the relations between characteristic time and model parameters. We not only present for the first time the derivation of the SNP model in the frequency domain, but also propose two new models — an LNP model and an nSNP model. In contrast to the original SNP model, which is limited to the case of small mobility contrasts, our models are valid for a wide range of mobility contrasts.

Our results confirm the quadratic dependence of the characteristic time on pore lengths. The apparent contradiction between earlier studies regarding the influence of the two pore types on the characteristic time is resolved. The characteristic time results to be dominated by the length of the narrow pore in the limiting case of SNP and by the length of the wide pore in the case of LNP. A physical explanation of the observed coexistence of two different characteristic time regimes is also suggested, as the diffusion relaxation seems to be dominated by the shorter of both possible characteristic times.

Beside the quadratic dependence of the characteristic time on the pore length, which was confirmed by all three models, our results suggest that pore throat diameter and pore fluid chemistry, which are parameterized in terms of ionic mobilities, play an important role in the determination of the characteristic time.

Our nSNP model is equal to a Cole-Cole model, if the frequency exponent of the latter equals 0.5. The LNP model derived here matches a Cole-Davidson model, again if the exponent is set to 0.5. Both analogies help to physically legitimize the application of generalized Cole-Cole models, which are widely used to fit spectral electrical impedance data. In addition, we were able to give a simple equivalent circuit representation of the Marshall-Madden impedance involving finite-length Warburg impedances that are typical for diffusion-dominated conduction mechanisms.

ACKNOWLEDGMENTS

We thank Andreas Weller and two anonymous reviewers for their helpful comments that improved the original version of the manuscript. This work was funded by the gebo research association. The authors wish to acknowledge the donors Baker Hughes and the Ministry of Science and Culture of Lower Saxony, Germany, for the financial support of this study.

APPENDIX A

DERIVATION OF TWO NEW IMPEDANCE MODELS

At first, the impedance formula after Marshall and Madden (1959) is rearranged to yield an appropriate form for the subsequent mathematical derivation. Therefore, the frequency-dependent terms X_i in of the Marshall-Madden impedance, equation 10, are expressed in terms of the newly defined time constants $\tau_i = \frac{\Delta L_i^2}{8D_{pi}t_{ni}}$

$$X_i = \left(\frac{i\omega}{2D_{pi}t_{ni}} \right)^{1/2} \frac{\Delta L_i}{2} = \sqrt{i\omega\tau_i}. \quad (\text{A-1})$$

Replacing the abbreviations A , B , and S_i by the corresponding full expressions given in equations 12, 13, and 14 and using the relation $1/\tanh(x) = \coth(x)$ leads to

$$Z(\omega) = \frac{\Delta L_1}{\mu_{p1}c_0F} \left[t_{p1} + \frac{D_{p1}\Delta L_2}{D_{p2}\Delta L_1} t_{p2} + \frac{(t_{n2}t_{p1} - t_{n1}t_{p2})^2}{t_{n1}\sqrt{i\omega\tau_1} \coth \sqrt{i\omega\tau_1} + \frac{D_{p2}\Delta L_1}{D_{p1}\Delta L_2} t_{n2}\sqrt{i\omega\tau_2} \coth \sqrt{i\omega\tau_2}} \right]. \quad (\text{A-2})$$

For the next conversion, we have applied the Nernst-Einstein relation (equation 6) and multiplied the whole term in brackets by $\Delta L_1/D_{p1}$

$$Z(\omega) = \frac{k_B T}{ec_0 F} \left[\frac{\Delta L_1 t_{p1}}{D_{p1}} + \frac{\Delta L_2 t_{p2}}{D_{p2}} + \frac{(t_{n2}t_{p1} - t_{n1}t_{p2})^2}{\frac{D_{p1}t_{n1}}{\Delta L_1} \sqrt{i\omega\tau_1} \coth \sqrt{i\omega\tau_1} + \frac{D_{p2}t_{n2}}{\Delta L_2} \sqrt{i\omega\tau_2} \coth \sqrt{i\omega\tau_2}} \right]. \quad (\text{A-3})$$

Using the definition of the transport numbers (equation 9) and the Nernst-Einstein relation (equation 6) for the first two terms and the definition of the time constant $\tau_i = \frac{\Delta L_i^2}{8D_{pi}t_{ni}}$ for the last term gives the following expression

$$Z(\omega) = \frac{k_B T}{ec_0 F} \left[\frac{\Delta L_1}{D_{p1} + D_{n1}} + \frac{\Delta L_2}{D_{p2} + D_{n2}} + \frac{8(t_{n2}t_{p1} - t_{n1}t_{p2})^2}{\frac{\Delta L_1}{\tau_1} \sqrt{i\omega\tau_1} \coth \sqrt{i\omega\tau_1} + \frac{\Delta L_2}{\tau_2} \sqrt{i\omega\tau_2} \coth \sqrt{i\omega\tau_2}} \right]. \quad (\text{A-4})$$

Obviously, the frequency dependence of this complex impedance is solely determined by the terms $\sqrt{i\omega\tau_i} \coth(\sqrt{i\omega\tau_i})$. In the limiting case of very small arguments $\omega\tau_i \ll 1$, these terms become 1. Thus, we can finally rewrite the Marshall-Madden impedance in terms of its DC limit Z_{DC} and its polarizability η_0

$$Z(\omega) = Z_{DC} \left[1 - \eta_0 \times \left(1 - \frac{\frac{\Delta L_1}{\tau_1} + \frac{\Delta L_2}{\tau_2}}{\frac{\Delta L_1}{\tau_1} \sqrt{i\omega\tau_1} \coth \sqrt{i\omega\tau_1} + \frac{\Delta L_2}{\tau_2} \sqrt{i\omega\tau_2} \coth \sqrt{i\omega\tau_2}} \right) \right], \quad (\text{A-5})$$

where the DC limit is given by

$$Z_{DC} = Z(\omega \rightarrow 0) = \frac{k_B T}{ec_0 F} \left[\frac{\Delta L_1}{D_{p1} + D_{n1}} + \frac{\Delta L_2}{D_{p2} + D_{n2}} + \frac{8(t_{n2}t_{p1} - t_{n1}t_{p2})^2}{\frac{\Delta L_1}{\tau_1} + \frac{\Delta L_2}{\tau_2}} \right], \quad (\text{A-6})$$

and the polarizability by

$$\eta_0 = \frac{k_B T}{ec_0 F} \frac{8(t_{n2}t_{p1} - t_{n1}t_{p2})^2}{\frac{\Delta L_1}{\tau_1} + \frac{\Delta L_2}{\tau_2}} \frac{1}{Z_{DC}}. \quad (\text{A-7})$$

Long narrow pore model

In this section, an approximate impedance model for the case of (1) relatively LNP $A < C/100$ and (2) approximately equal mobilities $\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$ is derived. Expanding the last term of equation A-5 by $\tau_1/\Delta L_1$ gives

$$Z(\omega) = Z_{DC} \left[1 - \eta_0 \times \left(1 - \frac{1 + \frac{\Delta L_2 \tau_1}{\Delta L_1 \tau_2}}{\sqrt{i\omega\tau_1} \coth \sqrt{i\omega\tau_1} + \frac{\Delta L_2 \tau_1}{\Delta L_1 \tau_2} \sqrt{i\omega\tau_2} \coth \sqrt{i\omega\tau_2}} \right) \right]. \quad (\text{A-8})$$

The ratio of the time constants

$$\frac{\tau_1}{\tau_2} = \frac{\Delta L_1^2}{8D_{p1}t_{n1}} \frac{8D_{p2}t_{n2}}{\Delta L_2^2} \quad (\text{A-9})$$

may be expressed in terms of the length ratio $A = \Delta L_1/\Delta L_2$ and the mobility contrast $C = \mu_{n1}/\mu_{n2}$. Condition (2) $\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$ yields $t_{n1} \approx 0.5$ and $D_{p1}/D_{p2} \approx 1$. For the anionic transport number in the active zone, we get

$$t_{n2} = \frac{\mu_{n2}}{\mu_{p2} + \mu_{n2}} = \frac{1}{\mu_{p2}/\mu_{n2} + 1} \approx \frac{1}{\mu_{n1}/\mu_{n2} + 1} = \frac{1}{C + 1}. \quad (\text{A-10})$$

Considering condition (1) $A < C/100$, the ratio of the time constants (equation A-9) may then be approximated as follows

$$\frac{\tau_1}{\tau_2} \approx \frac{2A^2}{C + 1} < \frac{2C}{100^2(1 + 1/C)}. \quad (\text{A-11})$$

In the case of moderate mobility contrasts (approximately $C \leq 100$), this ratio is very small ($\tau_1/\tau_2 \ll 1$ or $\tau_1 \ll \tau_2$). Moreover, the last term of the numerator in equation A-8 gives

$$\frac{\Delta L_2 \tau_1}{\Delta L_1 \tau_2} \approx \frac{2A}{C + 1} < \frac{2}{100(1 + 1/C)} \leq \frac{2}{100} \quad (\text{A-12})$$

and may therefore be neglected. In the denominator, the approximate ratio of the of the time constants (equation A-11) can be used for the substitution $\tau_2 \approx \tau_1(C + 1)/(2A^2)$

$$Z(\omega) \approx Z_{\text{DC}} \left[1 - \eta_0 \times \left(1 - \frac{1}{\sqrt{i\omega\tau_1} (\coth \sqrt{i\omega\tau_1} + (\frac{2}{C+1})^{1/2} \coth \sqrt{i\omega\tau_2})} \right) \right]. \quad (\text{A-13})$$

Furthermore, the magnitude of the imaginary part of $\coth(\sqrt{i\omega\tau_i})$ decreases at a sufficient rate with growing arguments $\omega\tau_i$, so that in conjunction with $\tau_1 \ll \tau_2$ we have $|\text{Im}(\coth \sqrt{i\omega\tau_1})| > |\text{Im}(\coth \sqrt{i\omega\tau_2})|$. For the real part, this relation does not apply in a similarly rigorous manner, but still holds $|\text{Re}(\coth \sqrt{i\omega\tau_1})| \geq |\text{Re}(\coth \sqrt{i\omega\tau_2})|$. As the mobility contrast is assumed to be $C \geq 1$, the factor $(2/(C+1))^{1/2}$ is smaller than 1. Neglecting the last term in the denominator of equation A-13, this leads to our new LNP model

$$Z_{\text{LNP}}(\omega) = Z_{\text{DC}} \left[1 - \eta_0 \left(1 - \frac{\tanh \sqrt{i\omega\tau_1}}{\sqrt{i\omega\tau_1}} \right) \right]. \quad (\text{A-14})$$

Although the derivation of this expression requires the additional assumption of moderate mobility contrasts ($C \leq 100$), it can be shown in a similar fashion that it holds for very pronounced mobility contrasts ($C \gg 100$) as well.

The characteristic time of the impedance model (equation A-14) is not equal to the time constant τ_1 , but can be calculated as the inverse of the characteristic frequency $\omega_c = 1/\tau_{\text{LNP}}$, where the phase maximum is observed. For $\omega = \omega_c$, the derivative of the phase (inverse tangent of the ratio of imaginary and real part of the impedance) with respect to the frequency is set to zero. This leads to a relation between ω_c , τ_1 , and η_0 , which cannot be solved explicitly for ω_c . Here, it is solved for η_0

$$\eta_0(\omega_c \tau_1) = 1 + 2 \times \frac{-\cot \sqrt{2\omega_c \tau_1} + \coth \sqrt{2\omega_c \tau_1}}{2\sqrt{2\omega_c \tau_1} - \frac{\cosh \sqrt{2\omega_c \tau_1}}{\sinh \sqrt{2\omega_c \tau_1}} + \frac{\cos \sqrt{2\omega_c \tau_1}}{\sinh \sqrt{2\omega_c \tau_1}} + \cot \sqrt{2\omega_c \tau_1} - \coth \sqrt{2\omega_c \tau_1}}. \quad (\text{A-15})$$

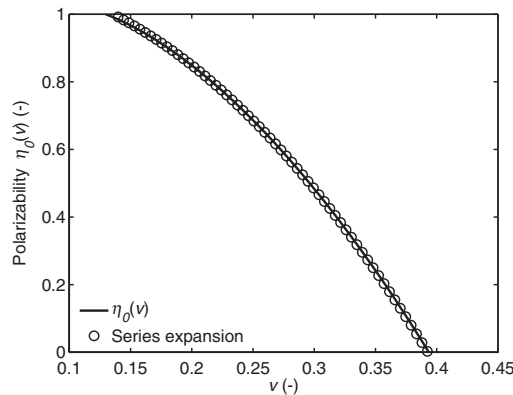


Figure A-1. The polarizability η_0 as a function of $v = 1/(\omega_c \tau_1)$ (black solid line) according to equation A-15 and the second-order approximation about $v = 0.25$ according to equation A-16 (gray dashed line) match very well within the range $0 \leq \eta_0 \leq 1$.

We are aiming at a relation $\tau_c = v(\eta_0) \cdot \tau_1$, where v is a dimensionless factor. For this purpose, $\sqrt{2\omega_c \tau_1}$ is expressed as $\sqrt{2/v}$. Thereafter, the function $\eta_0(v)$ is expanded in a Taylor series about $v = 0.25$, which gives a good fit (see Figure A-1). The numerical expansion yields

$$\eta_0(v) \approx \alpha + \beta(v - 0.25) + \gamma(v - 0.25)^2 \quad (\text{A-16})$$

with the coefficients

$$\alpha = 0.686272, \quad \beta = -3.65885 \quad \text{and} \quad \gamma = -8.03007. \quad (\text{A-17})$$

To finally yield the desired relation, the quadratic equation A-16 is solved for v .

$$v(\eta_0) = -\frac{\beta - 0.5\gamma}{2\gamma} \pm \left[\left(\frac{\beta - 0.5\gamma}{2\gamma} \right)^2 - \frac{\alpha - 0.25\beta + 0.0625\gamma - \eta_0}{\gamma} \right]^{1/2}. \quad (\text{A-18})$$

As only values $\eta_0 \geq 0$ are meaningful from a physical point of view, we are left with the positive root. The sought-after factor is given by

$$v(\eta_0) \approx 0.022178 + (0.137366 - 0.124532\eta_0)^{1/2}. \quad (\text{A-19})$$

In summary, the characteristic time of the LNP model, equation A-14, can be expressed in terms of the factor v as follows

$$\tau_{\text{LNP}} \approx [0.022178 + (0.137366 - 0.124532\eta_0)^{1/2}] \tau_1. \quad (\text{A-20})$$

For small polarizabilities $\eta_0 \ll 1$, this characteristic time reduces to $\tau_{\text{LNP}} \approx 0.4\tau_1$.

New short narrow pore model

In a similar fashion, an approximate impedance model can be found for the case of (1) relatively SNP $A \gg 5C$, (2) approximately equal mobilities $\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$, and (3) reduced anion mobilities in the active zone $\mu_{n1} > \mu_{n2}$. Further below, we will see that in the targeted frequency range ($\omega \approx \omega_c$) these three conditions imply $\omega\tau_1 \gg 1$ and $\omega\tau_2 < 1$. Using the latter relations, $\sqrt{i\omega\tau_1} \coth \sqrt{i\omega\tau_1}$ becomes $\sqrt{i\omega\tau_1}$ and $\sqrt{i\omega\tau_2} \coth \sqrt{i\omega\tau_2}$ may be approximated by 1. As a result, the Marshall-Madden impedance, equation A-5, becomes

$$Z(\omega) \approx Z_{\text{DC}} \left[1 - \eta_0 \left(1 - \frac{\frac{\Delta L_1}{\tau_1} + \frac{\Delta L_2}{\tau_2}}{\sqrt{i\omega\tau_1} + \frac{\Delta L_2}{\tau_2}} \right) \right]. \quad (\text{A-21})$$

Expansion of the last term of this equation by $\tau_2/\Delta L_2$ gives

$$Z(\omega) \approx Z_{\text{DC}} \left[1 - \eta_0 \left(1 - \frac{\frac{\Delta L_1 \tau_2}{\Delta L_2 \tau_1} + 1}{\frac{\Delta L_1 \tau_2}{\Delta L_2 \tau_1} \sqrt{i\omega\tau_1} + 1} \right) \right]. \quad (\text{A-22})$$

As before, condition (2) yields $t_{n1} \approx 0.5$, $D_{p1}/D_{p2} \approx 1$, and $t_{n2} \approx 1/(C+1)$ (see equation A-10). In conjunction with condition (1) $A \gg 5C$, the enumerator term becomes

$$\frac{\Delta L_1 \tau_2}{\Delta L_2 \tau_1} = \frac{\Delta L_1}{\Delta L_2} \frac{\Delta L_2^2}{8D_{p2} t_{n2}} \frac{8D_{p1} t_{n1}}{\Delta L_1^2} \approx \frac{0.5(C+1)}{A} \ll 1, \quad (\text{A-23})$$

and may therefore be neglected. After the introduction of a modified time constant

$$\tau' = \tau_2 \frac{D_{p1} t_{n1}}{D_{p2} t_{n2}}, \quad (\text{A-24})$$

equation A-22 reduces to our nSNP model given by

$$Z_{\text{nSNP}}(\omega) = Z_{\text{DC}} \left[1 - \eta_0 \left(1 - \frac{1}{1 + \sqrt{i\omega\tau'}} \right) \right]. \quad (\text{A-25})$$

To calculate the characteristic time τ_{nSNP} , the phase of the impedance, equation A-25, is differentiated with respect to the frequency and set to zero. The resulting characteristic frequency ω_c , where the phase maximum is observed is

$$\omega_c = \frac{1}{\tau'(1 - \eta_0)}. \quad (\text{A-26})$$

The characteristic time of the nSNP model may be expressed in terms of the polarizability η_0 and the modified time constant τ' .

$$\tau_{\text{nSNP}} = \frac{1}{\omega_c} = \tau'(1 - \eta_0). \quad (\text{A-27})$$

For small polarizabilities $\eta_0 \ll 0$, the characteristic time reduces to $\tau_{\text{nSNP}} \approx \tau'$.

Finally, the initially claimed implication of $\omega\tau_1 \gg 1$ and $\omega\tau_2 < 1$ from conditions (1), (2), and (3) can be demonstrated in the proximity of ω_c , where it should hold. For this purpose, we set $\omega = \omega_c$ and approximate ω_c by $1/\tau'$. With the modified time constant, equation A-24, this gives

$$\omega_c \tau_1 \approx \frac{\tau_1 D_{p2} t_{n2}}{\tau_2 D_{p1} t_{n1}}. \quad (\text{A-28})$$

As we already know, condition (2) $\mu_{p1} \approx \mu_{p2} \approx \mu_{n1}$ yields $t_{n1} \approx 0.5$ and $D_{p2}/D_{p1} \approx 1$. In addition, we have $t_{n2} \approx 1/(C+1)$ (see equation A-10). Substituting the time constant ratio, equation A-11, into equation A-28 and considering condition (1) $A \gg 5C$ gives

$$\omega_c \tau_1 \approx \left(\frac{2A}{C+1} \right)^2 \gg 1. \quad (\text{A-29})$$

With condition (3) $C = \mu_{n1}/\mu_{n2} > 1$ and the approximation $\omega_c \approx 1/\tau'$, the second implication may be easily shown

$$\omega_c \tau_2 \approx \frac{\tau_2}{\tau'} = \frac{D_{p2} t_{n2}}{D_{p1} t_{n1}} \approx \frac{2}{C+1} < 1. \quad (\text{A-30})$$

Titov's short narrow pore model

The original SNP model by Titov et al. (2002) can be derived in a similar manner. Here, the narrow pore is assumed to be absolutely shorter than the wide pore $A \gg 1$, unlike the nSNP model where the assumption was expressed relative to the mobility contrast. In addition, all mobilities are assumed to be approximately the same $\mu_{p1} \approx \mu_{p2} \approx \mu_{n1} \approx \mu_{n2}$ (i.e., $C \approx 1$). Expansion of the last term of equation A-5 by $\Delta L_2 \tau_1 / \Delta L_1^2$ gives

$$Z(\omega) \approx Z_{\text{DC}} \left[1 - \eta_0 \times \left(1 - \frac{\frac{1}{A} + 1}{\frac{1}{A} \sqrt{i\omega\tau_1} \coth \sqrt{i\omega\tau_1} + \sqrt{i\omega\tau_2} \coth \sqrt{i\omega\tau_2}} \right) \right]. \quad (\text{A-31})$$

Here, we have already considered the condition $\mu_{p1} \approx \mu_{p2} \approx \mu_{n1} \approx \mu_{n2}$ and the definition of the time constants $\tau_i = \Delta L_i^2 / (8D_{pi} t_{ni})$, leading to

$$\frac{\tau_1 \Delta L_2^2}{\tau_2 \Delta L_1^2} = \frac{D_{p2} t_{n2}}{D_{p1} t_{n1}} \approx 1. \quad (\text{A-32})$$

Similarly, one can show that $1/A\sqrt{i\omega\tau_1}$ can be approximated by $\sqrt{i\omega\tau_2}$. As $A \gg 1$, the term $1/A$ in the enumerator may be neglected and we get

$$Z(\omega) \approx Z_{\text{DC}} \left[1 - \eta_0 \times \left(1 - \frac{1}{\sqrt{i\omega\tau_2} (\coth \sqrt{i\omega\tau_1} + \coth \sqrt{i\omega\tau_2})} \right) \right]. \quad (\text{A-33})$$

This impedance model is supposed to hold in the proximity of $\omega_c \approx 1/\tau_2$. Therefore, the expression $\omega\tau_1$ can be approximated by $\omega_c \tau_1 \approx \frac{\tau_1}{\tau_2} = A^2 \gg 1$, so that $\coth \sqrt{i\omega_c \tau_1} \approx 1$. Altogether, we get the SNP impedance model

$$Z_{\text{SNP}}(\omega) = Z_{\text{DC}} \left[1 - \eta_0 \left(1 - \frac{1}{\sqrt{i\omega\tau_2} (1 + \coth \sqrt{i\omega\tau_2})} \right) \right]. \quad (\text{A-34})$$

When the cotangent function is expressed in terms of exponentials, a last rearrangement finally leads to the same expression as derived by Titov et al. (2002)

$$Z_{\text{SNP}}(\omega) = Z_{\text{DC}} \left[1 - \eta_0 \left(1 - \frac{1 - e^{-2\sqrt{i\omega\tau_2}}}{2\sqrt{i\omega\tau_2}} \right) \right]. \quad (\text{A-35})$$

Under the condition $\mu_{p2} \approx \mu_{n2}$, the time constant becomes

$$\tau_2 = \frac{\Delta L_2^2}{8D_{p2} t_{n2}} \approx \frac{\Delta L_2^2}{4D_{p2}}. \quad (\text{A-36})$$

To derive the corresponding characteristic time τ_{SNP} , the phase of the impedance, equation A-35, is differentiated with respect to the frequency. By setting this derivative to zero, we arrive at the

following equation, which relates the characteristic frequency ω_c with the polarizability η_0 and the time constant τ_2

$$\begin{aligned} & -\cos \sqrt{2\omega_c\tau_2} \left(\frac{1}{\eta_0} - 2 \right) 2e^{\sqrt{2\omega_c\tau_2}} - 2 \\ & - \sin \sqrt{2\omega_c\tau_2} \frac{1}{\eta_0} 2e^{\sqrt{2\omega_c\tau_2}} \\ & - 2 \sin \sqrt{2\omega_c\tau_2} \left(\frac{1}{\eta_0} - 1 \right) 2\sqrt{2\omega_c\tau_2} e^{\sqrt{2\omega_c\tau_2}} \\ & + 2e^{2\sqrt{2\omega_c\tau_2}} \left(\frac{1}{\eta_0} - 1 \right) = 0. \end{aligned} \quad (\text{A-37})$$

Because small mobility contrasts $C \approx 1$ are considered, only small polarizabilities are expected. Therefore, it is sufficient to analyze equation A-37 in the limit $\eta_0 \rightarrow 0$, where it becomes

$$\begin{aligned} & \cos \sqrt{2\omega_c\tau_2} + \sin \sqrt{2\omega_c\tau_2} + 2\sqrt{2\omega_c\tau_2} \sin \sqrt{2\omega_c\tau_2} \\ & - e^{\sqrt{2\omega_c\tau_2}} = 0. \end{aligned} \quad (\text{A-38})$$

Solving this equation numerically yields only one real-valued root $\omega_c\tau_2 \approx 0.892$. The approximate characteristic time τ_{SNP} is related to the time constant τ_2 as follows:

$$\tau_{\text{SNP}} = \frac{1}{\omega_c} \approx 1.12 \cdot \tau_2. \quad (\text{A-39})$$

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